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Heart Disease Prediction: Comparative Analysis of Classification Models

Introduction

This report presents a comprehensive analysis of machine learning models applied to heart disease prediction using the UCI Heart Disease dataset. The primary objective is to compare the performance of three distinct classification approaches: Bernoulli Naive Bayes (BernoulliNB), Gaussian Naive Bayes (GaussianNB), and Linear Regression with various regularization techniques.

The dataset, sourced from the UCI Machine Learning Repository, contains medical information about 303 patients with 14 attributes including both continuous clinical measurements (age, resting blood pressure, cholesterol, maximum heart rate, ST depression) and categorical health indicators (sex, chest pain type, fasting blood sugar, exercise-induced angina). The target variable indicates the presence of heart disease, defined as greater than 50% diameter narrowing in any major vessel (Detrano et al.).

Our analysis evaluates how each model responds to different feature sets and preprocessing approaches, with the goal of identifying the most effective model for predicting heart disease presence in patients. The models examined in this report are:

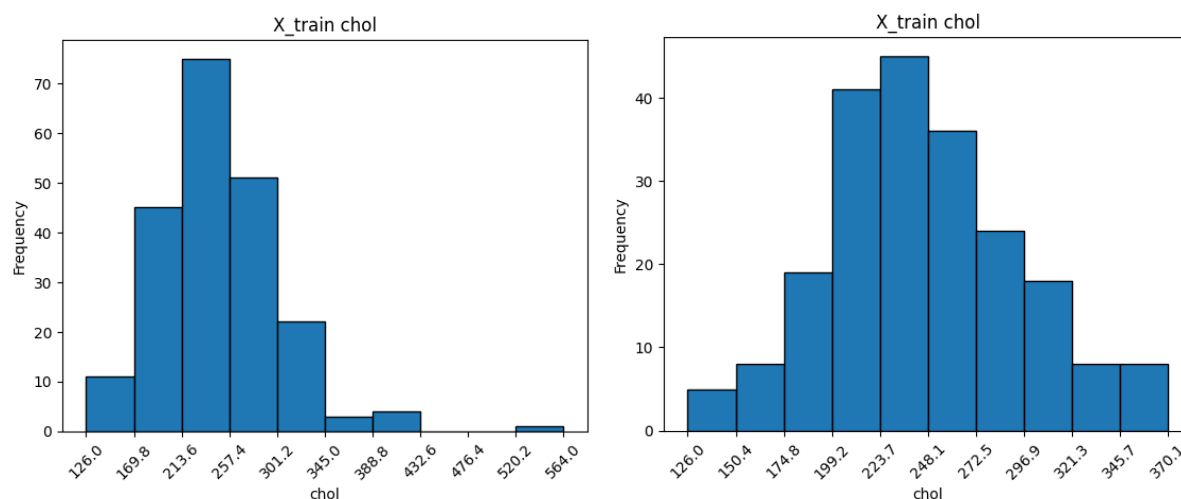
- **BernoulliNB:** A probabilistic classifier designed for binary/categorical features, tested with varying Laplace smoothing parameters ($\alpha = 0.00001, 0.01, 0.05, 1$)
- **GaussianNB:** A probabilistic classifier that assumes continuous features follow a Gaussian distribution
- **Linear Regression:** Implemented with three regularization approaches (regular, Ridge with $\alpha=1$, and LASSO with $\alpha=0$) to assess their impact on prediction performance and feature importance

1. Data Exploration

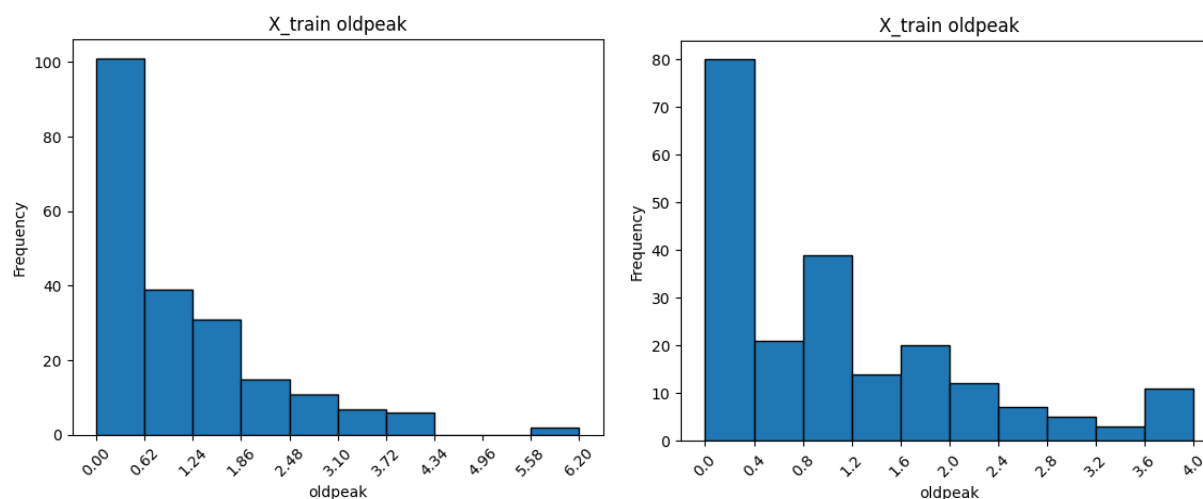
Initial data exploration began with examining hit rates and distributions for all variables using summary statistics and histogram visualizations. The hit rate analysis revealed that only two variables had missing values: 'ca' (number of major vessels, 98.7% complete) and 'thal' (thalassemia, 99.3% complete), with all other variables having 100% complete data.

Histogram analysis of continuous variables identified outliers requiring attention. One patient had a cholesterol value of 564 mg/dl (well above the 99th percentile of 406.7), and two patients

had oldpeak values exceeding 5.0 (5.6 and 6.2 compared to 99th percentile of 4.2). Capping was applied to these outliers using winsorizer to prevent them from disproportionately influencing model predictions later. Below is a side by side comparison of distributions with and without capping:



Before vs After capping Cholesterol



Before vs After capping Oldpeak

For missing values, 'ca' was imputed with the median value of 0 (which was also the mode), while 'thal', being categorical, was filled with -1 to create a distinct "unknown" category that could be one-hot encoded without losing information.

Correlation analysis of continuous variables showed that features were not closely correlated with each other (all correlations < 0.4 except age-thalach at -0.394), allowing all variables to be included in models without concern for competing signals. The variables most strongly correlated with the target were: ca (0.519), oldpeak (0.504), and thalach (-0.415).

	age	trestbps	chol	thalach	oldpeak	ca	num
age	1	0.284946	0.20895	-0.39381	0.203805	0.362605	0.222853
trestbps	0.284946	1	0.13012	-0.04535	0.189171	0.098773	0.157754
chol	0.20895	0.13012	1	-0.00343	0.046564	0.119	0.070909
thalach	-0.39381	-0.04535	-0.00343	1	-0.34309	-0.26425	-0.41504
oldpeak	0.203805	0.189171	0.046564	-0.34309	1	0.295832	0.504092
ca	0.362605	0.098773	0.119	-0.26425	0.295832	1	0.518909
num	0.222853	0.157754	0.070909	-0.41504	0.504092	0.518909	1

Correlation Matrix of Continuous Variables

2. Feature Engineering

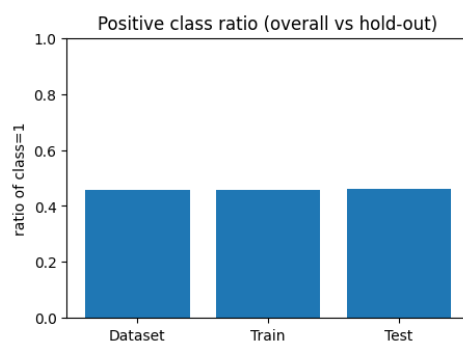
Feature engineering focused on transforming continuous variables into categorical representations suitable for BernoulliNB:

- Blood pressure was categorized as normal (≤ 130 mm Hg) or high (> 130 mm Hg) based on CDC guidelines.
- Age was grouped into life stages: young adulthood (< 40), middle adulthood (40-59), young old age (60-74), and old old age (≥ 75).
- Cholesterol was classified using John Hopkins guidelines as normal (< 200 mg/dL), borderline high (200-239 mg/dL), or high (≥ 240 mg/dL).
- For oldpeak, quantile-based binning starting at the 50th percentile (since 50% of values were 0) created 6 categories, while thalach was divided into deciles to capture heart rate ranges.

For data transformations before model training, one-hot encoding was applied to all categorical variables to create binary indicator features, and standardization (zero mean, unit variance) was applied to continuous variables for GaussianNB and Linear Regression models.

3. Model Training

The dataset was split 70-30 into training (212 observations) and testing (91 observations) sets using stratification with `random_state=5`. Stratification maintained consistent proportions of heart disease positive cases: 45.9% overall, 45.8% training, and 46.2% testing.



Each model was trained twice: first with features naturally suited to the model type (run1), and second with all features after applying appropriate transformations (run2). For BernoulliNB, the first run used only raw categorical variables, while the second included all feature-engineered categorical derivations. For GaussianNB and Linear Regression, the first run used only standardized continuous variables, while the second included continuous plus one-hot encoded categorical features.

Different hyperparameters were tested for each model type. BernoulliNB was evaluated with four Laplace smoothing values ($\alpha = 0.00001, 0.01, 0.05, 1$). Linear Regression was tested in three variants: regular, Ridge ($\alpha=1$), and LASSO ($\alpha=0$).

4. Model Evaluation

Performance by Model Type

BernoulliNB showed strong performance with feature-engineered variables, with $\alpha=1$ achieving the best results (accuracy 0.8132, AUC 0.8112) compared to raw categorical variables only (accuracy ranging from 0.7912 to 0.8022 depending on α). Smaller α values (0.01, 0.05, 0.00001) produced slightly lower but consistent results (accuracy 0.7912, AUC 0.7874), demonstrating that the smoothing parameter has a meaningful but moderate impact on this dataset's binary features.

GaussianNB showed dramatic improvement with all features (accuracy 0.6374, AUC 0.6548) versus continuous variables only (accuracy 0.4505, AUC 0.4881), demonstrating the critical importance of categorical information for heart disease prediction. However, this model significantly underperformed compared to both BernoulliNB and Linear Regression variants, indicating its Gaussian assumptions may not align well with this dataset's distributions.

Linear Regression variants showed minimal difference between continuous-only and all-features configurations, with both achieving similar accuracies. Ridge (0.8352 accuracy) produced best results, slightly outperforming regular Linear Regression (0.8242 accuracy).

Comparing all models, Ridge Regression (0.8352 accuracy, 0.9023 AUC) achieved the highest performance, followed closely by BernoulliNB with $\alpha=1$ (0.8132 accuracy, 0.8112 AUC), with GaussianNB (0.6374 accuracy, 0.6548 AUC) trailing significantly behind.

model	alpha	accuracy_run1	accuracy_run2	%Diff	roc/auc_run1	roc/auc_run2	%Diff
BernoulliNB	1	0.8022	0.8132	1%	0.8027	0.8112	1%
BernoulliNB	0.01	0.7912	0.7912	0%	0.7925	0.7874	-1%
BernoulliNB	0.05	0.8022	0.7912	-1%	0.8027	0.7874	-2%
BernoulliNB	0.00001	0.7912	0.7912	0%	0.7925	0.7874	-1%
GaussianNB	n/a	0.4505	0.6374	29%	0.4881	0.6548	25%
LinearReg	n/a	0.8242	0.8242	0%	0.9082	0.8989	-1%
LinearReg	0	0.5385	0.5385	0%	0.5	0.5	0%
LinearReg	1	0.8242	0.8352	1%	0.9096	0.9023	-1%

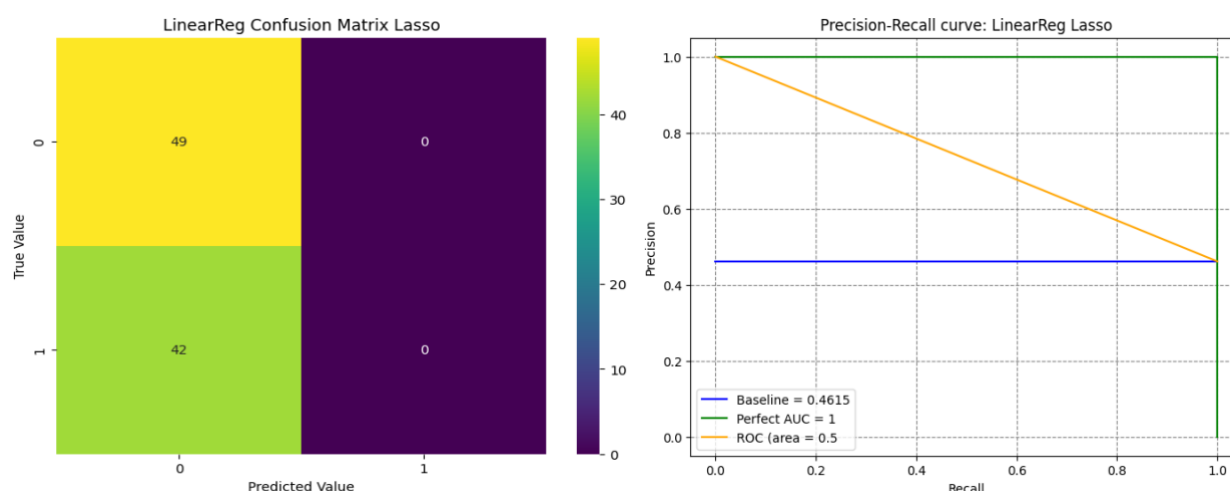
Linear Regression Regularization Comparison

Among the three Linear Regression variants, Ridge provided the best overall results across metrics, achieving accuracy of 0.8352, R^2 of 0.482436, MSE of 0.128625, and AUC of 0.9023.

model	MSE	MAE	R2
LinearReg_Ridge	0.1286	0.2809	0.4824
LinearReg_Regular LR	0.1323	0.2858	0.4676
LinearReg_Lasso	0.2485	0.4967	-0.0001

Metrics Table Summary for Linear Regression

LASSO produced the worst results, with accuracy of only 0.5385, near-zero R^2 (-0.000064), MSE of 0.248537, and AUC of 0.5, suggesting severe underfitting due to excessive regularization.

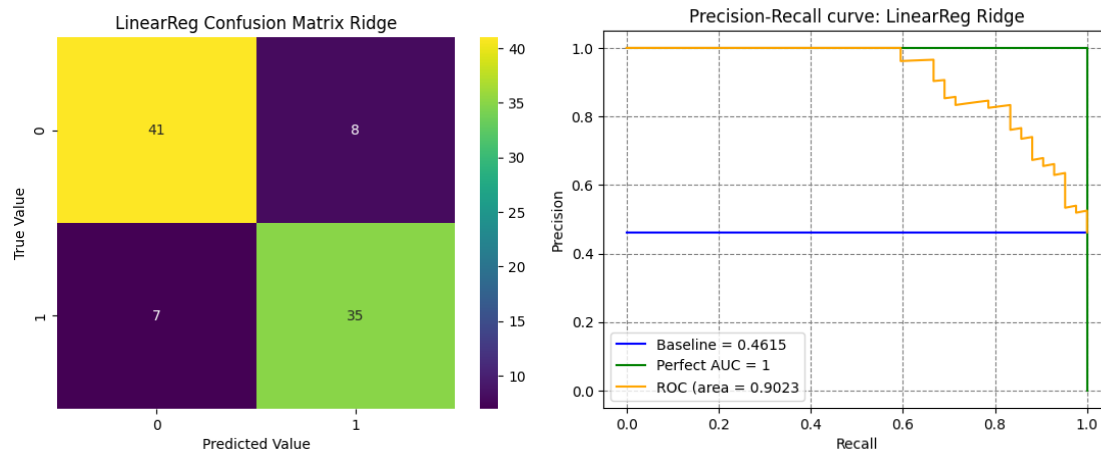


Examination of the coefficient table showed LASSO shrunk all variables to zero or near-zero, eliminating the predictive signal entirely. This was surprising given LASSO's theoretical strength in feature selection—some coefficients should have remained non-zero to identify important predictors. The extreme over-regularization, despite $\alpha=0$, rendered LASSO unsuitable for this application.

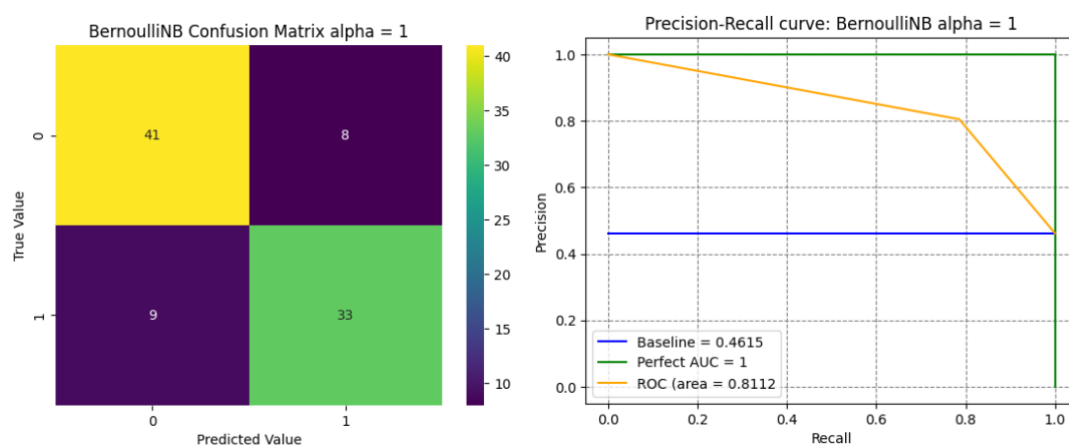
Best Model Selection

For fair comparison, we evaluated the best configuration from each model type: Ridge Regression with $\alpha=1$, BernoulliNB with $\alpha=1$, and GaussianNB with all features.

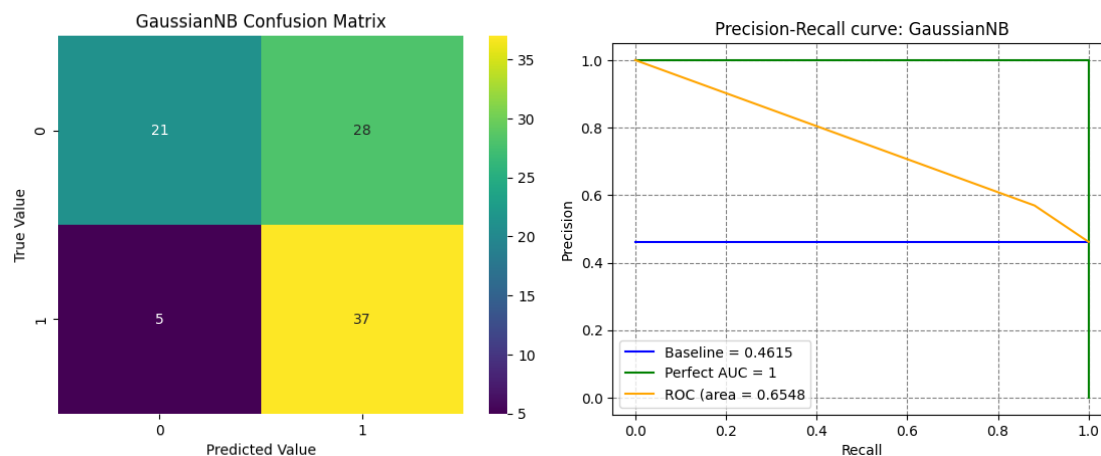
Ridge Regression emerged as the best overall model for this exercise with 0.8352 accuracy (76/91 correct) and excellent probability calibration reflected in AUC of 0.9023, demonstrating superior discriminative ability for this binary classification task.



BernoulliNB with $\alpha=1$ placed a close second with 0.8132 accuracy (74/91 correct) and strong discrimination with AUC of 0.8112, proving highly effective for this binary classification problem when paired with categorical feature engineering.



GaussianNB ranked third with 0.6374 accuracy (58/91 correct) and AUC of 0.6548, showing it struggles with this particular dataset and feature configuration.



Weighted average metrics confirmed these rankings. Ridge Regression achieved precision 0.836, recall 0.835, and F1-score 0.835. BernoulliNB with $\alpha=1$ achieved precision 0.813, recall 0.813, and F1-score 0.813. GaussianNB achieved precision 0.698, recall 0.637, and F1-score 0.621.

model_type	metrics	weighted avg
LinearReg_Ridge	precision	0.83561
LinearReg_Ridge	recall	0.83516
LinearReg_Ridge	f1-score	0.83528
BernoulliNB_alpha = 1	precision	0.81302
BernoulliNB_alpha = 1	recall	0.81319
BernoulliNB_alpha = 1	f1-score	0.81300
GaussianNB	precision	0.69763
GaussianNB	recall	0.63736
GaussianNB	f1-score	0.62073

6. Conclusion

This analysis demonstrated that proper data preprocessing order is critical for valid model evaluation. Ridge Regression with regularization ($\alpha=1$) achieved the best performance with 0.8352 accuracy and 0.9023 AUC, followed closely by BernoulliNB with $\alpha=1$, which benefited significantly from categorical feature engineering.

BernoulliNB, while slightly behind Ridge in accuracy (0.8132 vs 0.8352), still performed strongly with 0.8112 AUC, demonstrating that probabilistic classifiers can be highly effective for heart disease prediction when features are properly engineered into categorical representations.

LASSO's failure highlights the importance of careful hyperparameter selection and the potential dangers of extreme regularization, even with theoretically minimal α settings. Future work could explore cross-validated LASSO implementations to determine if appropriate regularization strength exists for this problem.

For clinical deployment, Ridge Regression with feature-engineered categorical variables ($\alpha=1$) is recommended due to its superior accuracy (0.8352) and probability calibration (AUC 0.9023), making it most reliable for clinical decision support. BernoulliNB with $\alpha=1$ serves as a strong alternative with 0.8132 accuracy and 0.8112 AUC.

Works Cited (Includes code references)

Detrano, R., et al. "International Application of a New Probability Algorithm for the Diagnosis of Coronary Artery Disease." *American Journal of Cardiology*, vol. 64, 1989, pp. 304-310.

"Heart Disease." *UCI Machine Learning Repository*, 2023, archive.ics.uci.edu/dataset/45/heart+disease. Accessed 6 Oct. 2025.

"High Blood Pressure." Centers for Disease Control and Prevention, 15 Jan. 2025, <https://www.cdc.gov/high-blood-pressure/index.html>. Accessed 6 Oct. 2025.

Hurley, Margaret Anne. "A Reference Relative Time-Scale as an Alternative to Chronological Age for Cohorts with Long Follow-Up." *Emerging Themes in Epidemiology*, vol. 12, 2015, doi:10.1186/s12982-015-0043-6. PubMed Central, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4684933/>. Accessed 6 Oct. 2025.

"Chronological Age." ScienceDirect Topics, Elsevier, <https://www.sciencedirect.com/topics/computer-science/chronological-age>. Accessed 6 Oct. 2025.

"Lipid Panel." Johns Hopkins Medicine, Johns Hopkins University, <https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/lipid-panel#:~:text=Normal:%20Less%20than%20150%20mg,14%20hours%20before%20this%20test.> Accessed 6 Oct. 2025.